Worksheet 4.3

Electron shells

Student:..... Class:.....

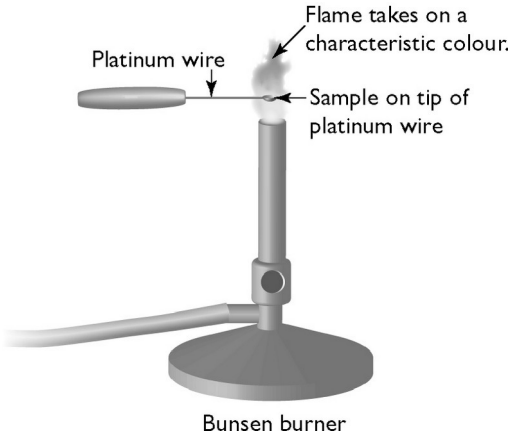
Electron configuration

1. Complete the following table.

Atom or ion	Proton number	Electron number	Electron configuration
Sodium atom (Na)	11		
Fluorine atom (F)		9	
Carbon atom (C)			2, 4
Calcium atom (Ca)			2, 8, 8, 2
Chloride ion (Cl^-)			2, 8, 8
Oxide ion (O^{2-})	8		
Magnesium ion (Mg^{2+})	12		2, 8
Aluminium ion (Al^{3+})	13		

Exciting the electrons

2. A student performed the experiment shown in the diagram below.



A sample of salt (sodium chloride) is placed on the tip of a platinum wire. When the sample is heated in a blue Bunsen flame the flame turns yellow.

Use the theory of electron shells to explain this observation.

Electron shell population

3. Each electron shell is represented by a number (n) as shown in the following table.

Shell number (n)	Maximum electron population
1	
2	
3	
4	

The maximum electron number in each shell is given by the formula:

$$\text{maximum electron number} = 2n^2$$

Complete the table by calculating the maximum electron population in each shell.

© John Wiley & Sons Australia, Ltd